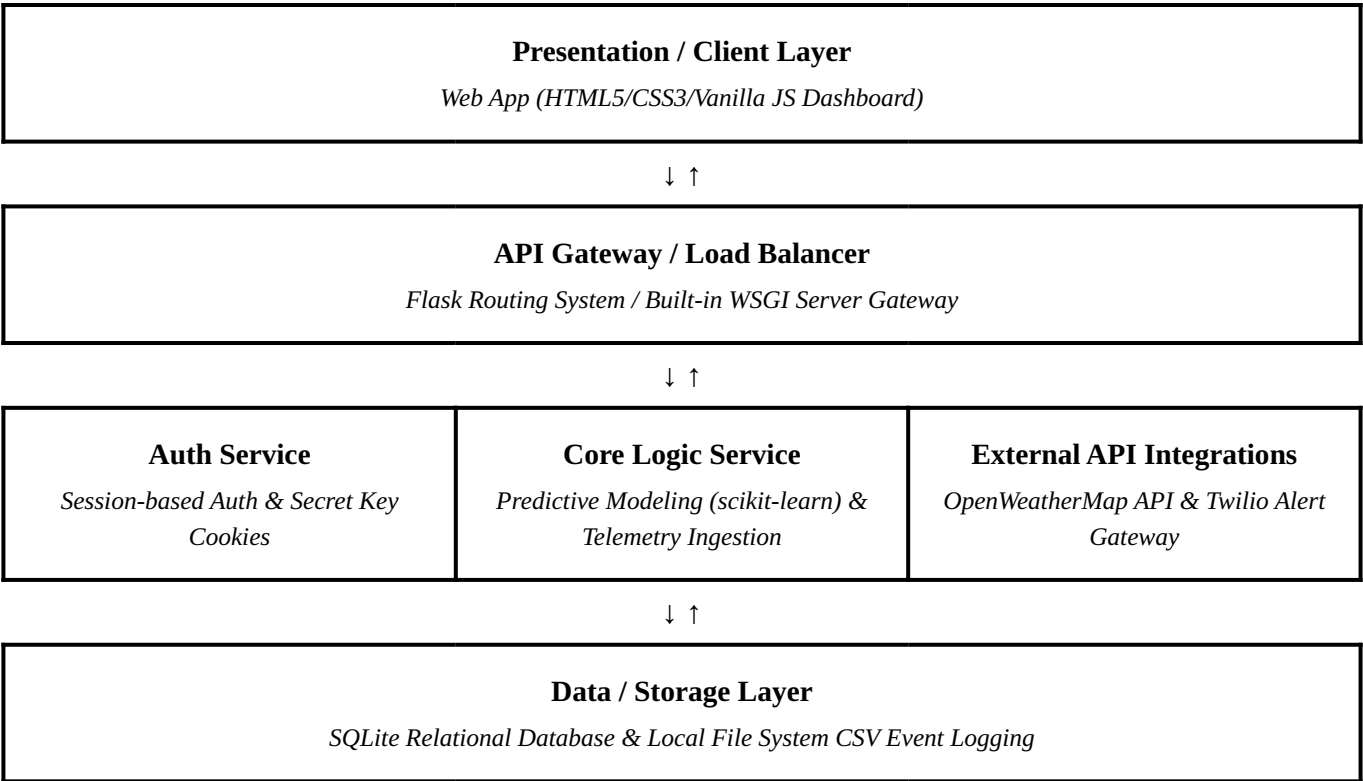


Solution Architecture

Date	1 July
Project Name	Rising Waters
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Renders real-time hazard map metrics, status tracking cards, alert threshold configurations, and operational summaries for emergency staff.	HTML5, CSS3, JavaScript (Vanilla ES6)
API Gateway	Manages inbound network routes, handles operational request mapping, and handles cross-origin routing configurations.	Flask Native Web Server Application Engine
Microservices	Processes incoming external rain metrics, executes the flood pattern matching algorithms, and handles Twilio SMS priority alert cascades.	Python (scikit-learn, pandas), Twilio SDK

Component Name	Description / Role in Architecture	Technologies Used
Database	Persistently saves sensor baseline metadata, timeline transaction events logs, and authorized personnel administrative access tables.	SQLite / Flat File CSV Format Data Stores